

CHAPTER – XII

APPENDIX

12.1 – Source Code

app.py

```
from flask import Flask,
render_template

app = Flask(__name__)

@app.route("/")
def home():
    return
    render_template("index.html")

@app.route("/about")
def about():
    return
    render_template("about.html")

@app.route("/dashboard01")
def dashboard01():
    return
    render_template("dashboard01.html")

@app.route("/dashboard02")
def dashboard02():
    return
    render_template("dashboard02.html")

@app.route("/story")
def story():
    return
    render_template("story.html")

if __name__ == "__main__":
    app.run(debug=True)
```

base.html

```
<!DOCTYPE html>
<html lang="en">

<head>
```

```
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width,
initial-scale=1">

<title>Strategic Product Placement
Analysis</title>

<link
href="https://cdn.jsdelivr.net/npm/
bootstrap@5.3.3/dist/css/bootstrap.
min.css" rel="stylesheet">

<link rel="stylesheet"
href="{{
url_for('static',filename='css/styl
e.css') }}">

</head>

<body>

<nav class="navbar navbar-expand-lg
navbar-dark custom-navbar">

<div class="container">

<a class="navbar-brand" href="/">
SPPA
</a>

<button class="navbar-toggler"
type="button"
data-bs-toggle="collapse"
data-bs-target="#nav">
<span class="navbar-toggler-
icon"></span>

</button>

<div class="collapse navbar-
collapse" id="nav">

<ul class="navbar-nav ms-auto">
```

```
<li class="nav-item">
  <a class="nav-link"
    href="/">Home</a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="/about">About</a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="/dashboard01">Dashboard
    01</a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="/dashboard02">Dashboard
    02</a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="/story">Story</a>
</li>
</ul>
```

```
</div>
```

```
</div>
```

```
</nav>
```

```
{% block content %}
{% endblock %}
```

```
<div class="footer">
  © 2026 Strategic Product
  Placement Analysis
</div>
```

```
<script
src="https://cdn.jsdelivr.net/npm/b
```

```
ootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
```

```
</body>  
</html>
```

Index.html

```
{% extends 'base.html' %}  
  
{% block content %}  
  
<section class="hero">  
  
<div>  
  
<h1>Strategic Product Placement  
Analysis</h1>  
  
<p>  
Analyze product positioning,  
customer behavior,  
sales trends and pricing  
strategies using Tableau.  
</p>  
  
<a href="/dashboard01" class="btn  
btn-light btn-lg mt-3">  
View Dashboard  
</a>  
  
</div>  
</section>  
  
<div class="container my-5">  
  
<div class="row g-4">  
  
<div class="col-md-3">  
<div class="dashboard-card">  
<h3>8</h3>  
<p>Visualizations</p>  
</div>  
</div>
```

```
<div class="col-md-3">
<div class="dashboard-card">
<h3>1500+</h3>
<p>Products</p>
</div>
</div>
```

```
<div class="col-md-3">
<div class="dashboard-card">
<h3>500+</h3>
<p>Customers</p>
</div>
</div>
```

```
<div class="col-md-3">
<div class="dashboard-card">
<h3>25</h3>
<p>Stores</p>
</div>
</div>
```

```
</div>
```

```
</div>
```

```
{% endblock %}
```

About.html

```
{% extends 'base.html' %}
```

```
{% block content %}
```

```
<div class="page-header">
<h1>About The Project</h1>
</div>
```

```
<div class="container my-5">
```

```
<div class="content-card">
```

```
<!-- Project Overview & Team Section
-->
```

```
<section class="py-5 bg-light">
  <div class="container">
```

```
<!-- Project Overview -->
<div class="text-center mb-
5">
    <h2 class="fw-bold text-
primary">My Project Overview</h2>
    <p class="lead">
        Strategic Product
Placement Analysis is a data-driven
project
        designed to identify
the most effective product placement
strategies within
retail environments. By analyzing
sales
        volume, customer
demographics, foot traffic, product
position,
        pricing, promotions,
and competitor pricing, the project
helps
        businesses optimize
product visibility and maximize
revenue.
    </p>
</div>

<!-- Team Section -->
<div class="text-center mb-
4">
    <h2 class="fw-
bold">Project Team</h2>
    <p class="text-
muted">Meet the members behind this
project</p>
</div>
<div class="row g-4">
    <!-- Member 1 -->
    <div class="col-lg-3 col-
md-6">
        <div class="card
shadow border-0 h-100 text-center">
            <div class="card-
body">
```

```

<h5
class="card-title">PASUPULETI PAVAN
KUMAR</h5>
<p
class="text-primary fw-bold">Team
Leader</p>
<p
class="card-text">
Responsible for project planning,
coordination,
and
overall project management.
</p>
</div>
</div>
</div>
<!-- Member 2 -->
<div class="col-lg-3 col-
md-6">
<div class="card
shadow border-0 h-100 text-center">
<div class="card-
body">

<h5
class="card-title">VANKADARI
PUNESH</h5>
<p
class="text-primary fw-bold">Data
Analyst</p>
<p
class="card-text">
Analyzes
```

datasets and creates insights for product placement strategies.

```
        </p>
      </div>
    </div>
  </div>

  <!-- Member 3 -->
  <div class="col-lg-3 col-
md-6">
    <div class="card
shadow border-0 h-100 text-center">
      <div class="card-
body">
        
        <h5
class="card-title">PALA CHANDRA
MOULI</h5>
        <p
class="text-primary fw-
bold">Dashboard Designer</p>
        <p
class="card-text">
          Develops
          interactive dashboards and
          visualizations
          using
          Tableau.
        </p>
      </div>
    </div>
  </div>
</div>
<!-- Member 4 -->
<div class="col-lg-3 col-
md-6">
  <div class="card
shadow border-0 h-100 text-center">
    <div class="card-
body">
```

```
<p  
class="card-text">  
    Prepares  
project reports, presentations, and  
technical  
documentation.  
</p>  
</div>  
</div>  
</div>  
</div>  
</section>  
</div>  
</div>
```

BRAVO


```
border-radius: 15px;
padding: 20px;
box-shadow: 0 4px 20px rgba(0,0,0,0.1);
overflow: hidden;
margin-top: 20px;
}

.tableauPlaceholder {
width: 100%;
}

.tableauViz {
width: 100% !important;
min-height: 850px;
}

@media (max-width: 992px) {
.tableauViz {
min-height: 700px;
}
}

@media (max-width: 768px) {
.tableauViz {
min-height: 600px;
}
}

@media (max-width: 576px) {
.tableauViz {
min-height: 500px;
}
}
</style>
<nav class="navbar navbar-expand-lg navbar-dark custom-navbar">

<div class="container">

<a class="navbar-brand" href="/">
SPPA
</a>

<button class="navbar-toggler"
type="button"
data-bs-toggle="collapse"
data-bs-target="#nav">
```

```
<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="nav">

<ul class="navbar-nav ms-auto">

<li class="nav-item">
<a class="nav-link" href="/">Home</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/about">About</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard01">Dashboard 01</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/dashboard02">Dashboard 02</a>
</li>

<li class="nav-item">
<a class="nav-link" href="/story">Story</a>
</li>

</ul>

</div>

</div>

</nav>

<div class="dashboard-container">
  <div class='tableauPlaceholder' id='tableauViz'>
    <noscript>
      <a href='#'>
        <img alt='Dashboard'

src='https://public.tableau.com/static/images/St/StrategicProductPlac
ementAnalysis-DASHBOARD01/StrategicProductPlacementAnalysis-
```

```
DashBoard1/1_rss.png'
    style='border:none' />
  </a>
</noscript>

  <object class='tableauViz'>
    <param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' />
    <param name='embed_code_version' value='3' />
    <param name='site_root' value='' />
    <param name='name'
      value='StrategicProductPlacementAnalysis-
DASHBOARD01/StrategicProductPlacementAnalysis-DashBoard1' />
    <param name='tabs' value='no' />
    <param name='toolbar' value='yes' />
    <param name='animate_transition' value='yes' />
    <param name='display_static_image' value='yes' />
    <param name='display_spinner' value='yes' />
    <param name='display_overlay' value='yes' />
    <param name='display_count' value='yes' />
    <param name='language' value='en-GB' />
  </object>
</div>
</div>

<script type="text/javascript">
  var divElement = document.getElementById('tableauViz');
  var vizElement = divElement.getElementsByTagName('object')[0];

  vizElement.style.width = '100%';
  vizElement.style.height = '850px';

  var scriptElement = document.createElement('script');
  scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
  vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>
```

Dashboard02

```
{% extends 'base.html' %}

{% block content %}

<div class="page-header">
```

```
<h1>Dashboard 02</h1>
</div>

<div class="container-fluid my-5">

  <div class="content-card">

    <div class="dashboard-container">

      <div class="tableauPlaceholder" id="tableauViz2">

        <noscript>
          <a href="#">
            
          </a>
        </noscript>

        <object class="tableauViz">
          <param name="host_url"
value="https%3A%2F%2Fpublic.tableau.com%2F" />
          <param name="embed_code_version" value="3" />
          <param name="site_root" value="" />
          <param name="name"
value="StrategicProductPlacementAnalysis-
DASHBOARD02/StrategicProductPlacementAnalysis-DashBoard2" />
          <param name="tabs" value="no" />
          <param name="toolbar" value="yes" />
          <param name="animate_transition" value="yes" />
          <param name="display_static_image" value="yes" />
          <param name="display_spinner" value="yes" />
          <param name="display_overlay" value="yes" />
          <param name="display_count" value="yes" />
          <param name="language" value="en-GB" />
        </object>

      </div>

    </div>

  </div>
```

```
</div>

</div>

<script type="text/javascript">
  var divElement = document.getElementById('tableauViz2');
  var vizElement = divElement.getElementsByTagName('object')[0];

  vizElement.style.width = '100%';
  vizElement.style.height = '900px';

  var scriptElement = document.createElement('script');
  scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
  vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

{% endblock %}
```

Story.html

```
{% extends 'base.html' %}

{% block content %}

<div class="page-header">
  <h1>Project Story</h1>
</div>

<div class="container-fluid my-5">

  <div class="content-card">

    <h3 class="text-center mb-4">
      Strategic Product Placement Analysis – Story
    </h3>

    <div class="row mb-4">

      <div class="col-md-12">

        <div class="story-card">
```

```
</div>

</div>

</div>

<div class="dashboard-container">

  <div class="tableauPlaceholder" id="tableauStory">

    <noscript>
      <a href="#">
        
      </a>
    </noscript>

    <object class="tableauViz">
      <param name="host_url"
value="https%3A%2F%2Fpublic.tableau.com%2F" />
      <param name="embed_code_version" value="3" />
      <param name="site_root" value="" />
      <param name="name"
value="StrategicProductPlacementAnalysis-STORY/Story1" />
      <param name="tabs" value="no" />
      <param name="toolbar" value="yes" />
      <param name="animate_transition" value="yes" />
      <param name="display_static_image" value="yes" />
      <param name="display_spinner" value="yes" />
      <param name="display_overlay" value="yes" />
      <param name="display_count" value="yes" />
      <param name="language" value="en-GB" />
    </object>

  </div>

</div>

</div>
```

```
</div>
```

```
<script type="text/javascript">
  var divElement = document.getElementById('tableauStory');
  var vizElement = divElement.getElementsByTagName('object')[0];

  vizElement.style.width = '100%';
  vizElement.style.height = '950px';

  var scriptElement = document.createElement('script');
  scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
  vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

{% endblock %}
```

Style.css

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
}

body{
  font-family:'Segoe UI',sans-serif;
  background:#f5f7fb;
}

/* Navbar */

.custom-navbar{
  background:linear-gradient(90deg,#0f172a,#1e3a8a);
}

.navbar-brand{
  font-weight:700;
  font-size:1.3rem;
}

.nav-link{
  color:white !important;
  margin-left:15px;
```

```
    font-weight:500;
}

.nav-link:hover{
    color:#facc15 !important;
}

/* Hero */

.hero{
    min-height:85vh;
    background:linear-gradient(135deg,#2563eb,#7c3aed);
    color:white;
    display:flex;
    align-items:center;
    justify-content:center;
    text-align:center;
    padding:40px;
}

.hero h1{
    font-size:3.5rem;
    font-weight:bold;
}

.hero p{
    font-size:1.2rem;
    margin-top:15px;
}

/* Cards */

.dashboard-card{
    background:white;
    border-radius:20px;
    padding:30px;
    text-align:center;
    box-shadow:0 4px 20px rgba(0,0,0,0.1);
    transition:0.3s;
}

.dashboard-card:hover{
    transform:translateY(-10px);
}

.dashboard-card h3{
```

```
    color:#2563eb;
    font-weight:bold;
}

/* Content */

.page-header{
    background:linear-gradient(135deg,#2563eb,#7c3aed);
    color:white;
    padding:60px 0;
    text-align:center;
}

.content-card{
    background:white;
    border-radius:20px;
    padding:30px;
    box-shadow:0 4px 20px rgba(0,0,0,0.08);
}

.dashboard-placeholder{
    height:500px;
    border:3px dashed #2563eb;
    border-radius:15px;
    display:flex;
    justify-content:center;
    align-items:center;
    background:#eef4ff;
    font-size:24px;
    font-weight:bold;
    color:#2563eb;
}

.footer{
    background:#0f172a;
    color:white;
    text-align:center;
    padding:20px;
    margin-top:50px;
}
```

